Contents

1 Games of Complete Information

1.1 Static Games of Complete Information

1.1.1 Nash Equilibrium

We start with a classic story:

Prisoners' Dilemma

Two suspects are arrested and charged with a crime. The police lack sufficient evidence to convict the suspects, unless at least one confesses. The suspects are held in separate cells and told:

- If only one of you confesses and testifies against your partner, then the person who confesses will go free while the person who does not confess will be convicted and given a 20-year jail sentence.
- If both of you confess, then both will be convicted and sent to prison for 5 years.
- Finally, if neither of you confesses, then both of you will be convicted of a minor offence and sentenced to 1 year in jail .

What should the suspects do?

To investigate the situation, we can represent the prisoners' decisions using a matrix:

$$\begin{array}{c c} & \begin{array}{cc} \text{Holdout} & \text{Confess} \end{array} \\ \begin{array}{c} \text{Holdout} \\ \text{Confess} \end{array} & \begin{bmatrix} (-1,-1) & (-20,0) \\ (0,-20) & (-5,-5) \end{bmatrix} \end{array}.$$

With some observation, it is easy to see that no matter what they other prisoner chooses to do, the best decision for the prisoner is to confess. Note that in this representation, we represent a *strategy* with an ordered pair consisting of the decisions made by the two prisoners respectively. Extending this idea to a higher dimension, we see that every strategy of a game is formed by an ordered n-tuple, where each factor represents a decision taken by the corresponding player.

Definition 1.1.1 ▸ Strategy Space

In an n-player game, the **strategy space** of the i-th player, denoted by S_i, is the set of all possible decisions for the player.

In the above example, for each strategy (i, j), we then place the years of imprisonment, or the *payoff*, for the two prisoners following that strategy in the corresponding matrix entry. Intuitively, for an n-player game in general, we can assign a payoff value for each strategy in $\prod_{i=1}^{n} S_i$, the set of all strategies.

Definition 1.1.2 ▸ Payoff Function

Let S_i be the strategy space of the i-th player in an n-player game, the **payoff function** for the i-th player is a map $u_i \colon \prod_{j=1}^{n} S_j \to \mathbb{R}$.

This gives us a convenient way to represent a game quantitatively.

Definition 1.1.3 ▸ Normal-Form of Games

Let G be an n-player game. Let S_i be the strategy space and u_i be the payoff function of the i-th player, then the **normal-form**, or **strategic form** of G is defined as

$$G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}.$$

It is important to note that the payoff a player depends not only on his own action but also on the actions of others. This inter-dependence is the essence of games.

Remark. For 2-player games, the payoff can be represented in a matrix.

Given any game, our goal is to derive the optimal strategy for the players. Before that, we shall discuss some criteria under which a strategy cannot be optimal. For the sake of notational simplicity, we denote

$$s_{-i} := (s_1, \cdots, s_{i-1}, s_{i+1}, \cdots, s_n)$$

and

$$s := (s_1, s_2, \cdots, s_n) = (s_i, s_{-i}).$$

Analogously, we denote

$$S_{-i} := \prod_{j=1}^{i-1} S_j \times \prod_{j=i+1}^{n} S_j.$$

Definition 1.1.4 ▸ Dominated and Dominant Strategies

Let $G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}$ be a normal-form game and $s_i, s_i' \in S_i$ be two strategies for the i-th player. We say that s_i is **strictly dominated** by s_i' or that s_i' is **strictly dominant** over s_i if

$$u_i(s_i, \boldsymbol{s}_{-i}) < u_i(s_i', \boldsymbol{s}_{-i}) \quad \text{for all } \boldsymbol{s}_{-i} \in S_{-i}.$$

Remark. In other words, s_i is strictly dominated by s_i' if and only if it is a strictly worse decision no matter what the other players decide to do.

If a strategy s is strictly dominated by any other strategy, then it is not a rational strategy because it can never be optimal.

Assuming that in an n-player game, every player knows exactly what the strategies are and their respective payoffs, then we can proceed with the following reasoning:

Technique 1.1.5 ▸ Iterated Elimination of Strictly Dominated Strategies (IESDS)

- Suppose that s_{i_0} is strictly dominated by any other strategy of player i, then if the player is rational, the strategy will never be played.
- Therefore, we can reduce the strategy space to $\left\{\boldsymbol{s} \in \prod_{i=1}^{n} S_i : s_i \neq s_{i_0}\right\}$.
- Apply the same reasoning on the reduced strategy space iteratively. If at each iteration there exists such a strictly dominated strategy, then we can successfully obtain the optimal strategy for the n-players.

However, in most games, we might not be able to find such a strictly dominated strategy. Therefore, we have to think from a conditional context: given a set of strategies played by the other players, what is the *best response* for the current player?

Definition 1.1.6 ▸ Best-Response Correspondence

Let $G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}$ be a normal-form game. The **best-response correspondence** for player i is a map $R_i : S_{-i} \to \mathcal{P}(S_i)$ defined by

$$R_i(\boldsymbol{s}_{-i}) := \operatorname*{argmax}_{s_i \in S_i} u_i(s_i, \boldsymbol{s}_{-i}).$$

Now, given $\boldsymbol{s}_{-i}$, the actions played by the other players, $R_i(\boldsymbol{s}_{-i})$ tells us what the best responses are for player i for his own profit. In other words, player i has **no incentive** to choose any strategy in $S_i \setminus R_i(\boldsymbol{s}_{-i})$. If such a situation holds for all the players, then it is clear that the game has reached an *equilibrium* where no player is incentivised to deviated from the current strategy.

It is clear that such equilibriums occur exactly when the strategies of all players are the best responses of one another.

Definition 1.1.7 ▸ Nash Equilibrium

Let $G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}$ be a normal-form n-player game, then the strategies $(s_1^*, s_2^*, \cdots, s_n^*)$ are a **Nash equilibrium** if

$$s_i^* \in R_i\left(\boldsymbol{s}_{-i}^*\right) \quad \text{for all } i = 1, 2, \cdots, n,$$

or equivalently,

$$u_i\left(s_i^*, \boldsymbol{s}_{-i}^*\right) = \max_{s_i \in S_i} u_i\left(s_i, \boldsymbol{s}_{-i}^*\right) \quad \text{for all } i = 1, 2, \cdots, n.$$

Now, define

$$G\left(R_i\right) := \left\{\left(s_i, \boldsymbol{s}_{-i}\right) : s_i \in R_i\left(\boldsymbol{s}_{-i}\right), \boldsymbol{s}_{-i} \in S_{-i}\right\}$$

as the **graph** of R_i, then one can check that $(s_1^*, s_2^*, \cdots, s_n^*)$ are a Nash equilibrium if and only if

$$\left(s_1^*, s_2^*, \cdots, s_n^*\right) \in \bigcap_{i=1}^{n} G\left(R_i\right).$$

Next, let us consider a simple case of bi-matrix games. Take any guess $(s_1, s_2) \in S_1 \times S_2$. We can compute $R_2(s_1)$ and $R_1(s_2)$, and (s_1, s_2) are a Nash equilibrium if and only if $s_1 \in R_1(s_2)$ and $s_2 \in R_2(s_1)$ by Definition 1.1.7.

Therefore, we can simply repeat this calculation for every pair $(s_1, s_2) \in S_1 \times S_2$ and deduce the set $G(R_1) \cap G(R_2)$, which consists of all Nash equilibria.

Proposition 1.1.8 ▸ Relation between Nash Equilibrium and IESDS

Let $G = (S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n)$ be an n-player normal-form game. If the strategies $(s_1^*, s_2^*, \cdots, s_n^*)$ are a Nash equilibrium, then each s_i^* cannot be eliminated by IESDS.

Proof. Suppose on contrary that there exists some s_i^* which is eliminated by IESDS. Without loss of generality, assume s_i^* to be the first in $(s_1^*, s_2^*, \cdots, s_n^*)$ to be eliminated. By Technique 1.1.5, there exists some $s_i' \in S_i$ with $s_i' \neq s_i^*$ such that

$$u_i\left(s_i^*, \boldsymbol{s}_{-i}\right) < u_i\left(s_i', \boldsymbol{s}_{-i}\right)$$

for all $\boldsymbol{s}_{-i} \in S_{-i}$. In particular, this means that

$$u_i\left(s_i^*, \boldsymbol{s}_{-i}^*\right) < u_i\left(s_i', \boldsymbol{s}_{-i}^*\right).$$

However, by Definition 1.1.7, we have $u_i(s_i^*, \boldsymbol{s}_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, \boldsymbol{s}_{-i}^*)$, which is a contradiction. □

Let S be the resultant strategy space obtained from IESDS and N be the set of all Nash equilibria of a game, then the above proposition essentially means that $N \subseteq S$.

Note that this does not mean that Proposition 1.1.8 establishes the **existence** of Nash equilibria for any game. However, we can indeed conclude a little bit about the uniqueness of Nash equilibrium under some circumstances.

Proposition 1.1.9 ▸ Uniqueness of Nash Equilibrium from IESDS

Let $G = (S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n)$ be an n-player normal-form game with finite strategy spaces. If $(s_1^*, s_2^*, \cdots, s_n^*) \in S$ are the only strategies which are not eliminated by IESDS, then it is the unique Nash equilibrium.

Proof. The uniqueness is obvious from Proposition 1.1.8, so it suffices to prove that the strategies $(s_1^*, s_2^*, \cdots, s_n^*)$ are indeed a Nash equilibrium. Suppose on contrary that it is not a Nash equilibrium, then there exists some s_i^* such that $s_i^* \notin R_i(\boldsymbol{s}_{-i}^*)$. Take some $b_i \in R_i(\boldsymbol{s}_{-i}^*)$, then

$$u_i(b_i, \boldsymbol{s}_{-i}^*) > u_i(s_i^*, \boldsymbol{s}_{-i}^*).$$

However, since b_i is eliminated, it must be strictly dominated by some $t_i \in S_i$ such that $t_i \neq b_i$. Therefore, we have

$$u_i(t_i, \boldsymbol{s}_{-i}^*) > u_i(b_i, \boldsymbol{s}_{-i}^*)$$

because $\boldsymbol{s}_{-i}^*$ are strategies which are not eliminated, but this is a contradiction since t_i is a better response than b_i. □

So far, in our analysis we have been assuming that the players in a game make decisions **deterministically**, i.e., the player always choose a single strategy as the response. However, in real-life scenarios, there could be some random factors involved in decision-making.

For example, consider a game where 2 players, A and B, choose which side of a coin to display. If both choose the same side, then B wins over A's coin. Otherwise, A wins over B's coin. In this case, the payoff matrix is

$$\begin{array}{c c} & \begin{array}{cc} \text{Head} & \text{Tail} \end{array} \\ \begin{array}{c} \text{Head} \\ \text{Tail} \end{array} & \begin{bmatrix} (-1,1) & (1,-1) \\ (1,-1) & (-1,1) \end{bmatrix} \end{array}.$$

One may check that this game does not have a Nash equilibrium because the best responses of the two players are always disjoint. Such a phenomenon occurs because we assume that both players move deterministically, whereas in reality, since each player's best move is **dependent on the opponent's move** but there is no prior information on the opponent's move before the player's move is made, there is uncertainty in the actual moves executed.

Therefore, we need to consider the probabilities of each strategy being executed.

Definition 1.1.10 ▸ Mixed Strategy

Let $G = (S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n)$ be an n-player normal-form game with finite strategy spaces. Suppose $S_i = \{s_{i1}, s_{i2}, \cdots, s_{ik}\}$, then each $s_{ij} \in S_i$ is a **pure strategy** for player i and a probability distribution $p_i := (p_{i1}, p_{i2}, \cdots, p_{ik})$ is said to be a **mixed strategy** such that the probability of player i playing s_{ij} is p_{ij}.

As an example, we consider two players, i and j, with M and N pure strategies respectively. Let p_i and p_j be mixed strategies for i and j respectively, then the **expected payoff** of player i playing any pure strategy s_{ik} is

$$v_i\left(s_{ik}, p_j\right) := \sum_{n=1}^{N} p_{jn} u_i\left(s_{ik}, s_{jn}\right).$$

With this, we can compute the **expected payoff** for the mixed strategy p_i as

$$\begin{aligned} v_i\left(p_i, p_j\right) &:= \sum_{m=1}^{M} p_{im} v_i\left(s_{im}, p_j\right) \\ &= \sum_{m=1}^{M} \sum_{n=1}^{N} p_{im} p_{jn} u_i\left(s_{im}, s_{jn}\right). \end{aligned}$$

Similarly, the expected payoff for p_j is

$$v_j\left(p_j, p_i\right) = \sum_{n=1}^{N} \sum_{m=1}^{M} p_{jn} p_{im} u_j\left(s_{jn}, s_{im}\right).$$

Now we can extend the notion of Nash equilibrium to mixed strategies using expected payoffs.

Definition 1.1.11 ▸ Nash Equilibrium for Mixed Strategies

Let $G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}$ be a normal-form n-player game, then the probability distributions $(p_1^*, p_2^*, \cdots, p_n^*)$ are a **Nash equilibrium** if

$$v_i\left(p_i^*, \boldsymbol{p}_{-i}^*\right) \geq v_i\left(p_i, \boldsymbol{p}_{-i}^*\right)$$

for any probability distribution p_i on S_i.

Consider a 2-player game where each player has exactly 2 strategies. Let $p_1 = (r, 1-r)$ and $p_2 = (q, 1-q)$ be mixed strategies for player 1 and 2 respectively, then based on the previous argument the expected payoff of player 1 is

$$v_1\left(p_1, p_2\right) = r v_1\left(s_{11}, p_2\right) + (1-r)\, v_1\left(s_{12}, p_2\right).$$

For each $q \in [0, 1]$, we can compute

$$r^*(q) := \operatorname*{argmax}_{0 \leq r \leq 1} v_1\left(p_1, p_2\right).$$

This is known as the *best response* of player 1 to the mixed strategy p_2. Observe that v_1 is a **convex combination**, so it is clear that

$$r^*(q) = \begin{cases} \{1\} & \text{if } v_1\left(s_{11}, p_2\right) > v_1\left(s_{12}, p_2\right) \\ \{0\} & \text{if } v_1\left(s_{11}, p_2\right) < v_1\left(s_{12}, p_2\right) \\ [0, 1] & \text{if } v_1\left(s_{11}, p_2\right) = v_1\left(s_{12}, p_2\right) \end{cases}.$$

The case of $q^*(r)$ is symmetric. Taking $r^*(q) \cap q^*(r)$ yields the Nash equilibrium involving mixed strategies.

If there are more than 2 strategies for a player, we should first eliminate strictly dominated strategies to reduce the strategy spaces. Intuitively, if a pure strategy s_{ik} is eliminated by IESDS, then any mixed strategy that is good enough probably should not favour playing the strategy. Indeed, we have the following result:

Proposition 1.1.12 ▸ Zero Probability of Strategies Eliminated by IESDS

Let $(p_1^*, p_2^*, \cdots, p_n^*)$ be any mixed strategy Nash equilibrium. If s_{ik} is eliminated by IESDS, then $p_{ik}^* = 0$.

Lastly, combining the two scenarios of Nash equilibria, the following theorem states formally the existence of Nash equilibria:

Theorem 1.1.13 ▸ Nash's Theorem

Let $G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}$ *be a normal-form* n*-player game with finite strategy spaces, then there exists at least one Nash equilibrium if mixed strategies are allowed.*

1.1.2 Economic Models

Suppose that there are two firms producing the same product. Let q_1 and q_2 be the quantity produced by the two firms respectively and c be the unit cost of production. Assuming that the demand is a linear function in price, then there exists positive constants q_0 and k such that

$$Q(P) = q_0 - kP,$$

where Q is the demand and P is the price. Suppose that the demand vanishes when the price reaches a.

Definition 1.1.14 ▸ Demand-Side Reservation Price

The **demand-side reservation price** is the lowest price at which the demand is 0.

Clearly, this means that $Q(a) = 0$ and so $q_0 = ka$. Therefore,

$$P(Q) = \begin{cases} 0 & \text{if } Q \geq ka \\ a - \frac{1}{k}Q & \text{otherwise} \end{cases},$$

Here our goal is to achieve *market clearing price*.

Definition 1.1.15 ▸ Market Clearing Price

The **market clearing price** is the price at which the demand is equal to the supply.

In order to achieve the market clearing price, the demand should always be equal to the supply, so the firm should produce such that $Q = q_1 + q_2$. This means that

$$P(Q) = \begin{cases} 0 & \text{if } Q \geq ka \\ a - \frac{1}{k}q_1 - \frac{1}{k}q_2 & \text{otherwise} \end{cases}$$

Define the profit by

$$\pi_i(q_1, q_2) = (P(Q) - c)q_i$$

for $i = 1, 2$, where c is the unit cost of production, then we have

$$\pi_i(q_1, q_2) = \begin{cases} \left(a - \frac{1}{k}q_1 - \frac{1}{k}q_2 - c\right) q_i & \text{if } Q < ka \\ -cq_1 & \text{if } Q \geq ka \end{cases}.$$

Definition 1.1.16 ▸ Cournot Model of Duopoly

Let $S_1, S_2 := [0, \infty)$ be possible supply quantities for a product produced by two firms. Assuming linear demand with demand-side reservation price a, then there exists some $k > 0$ such that

$$\pi_i(q_1, q_2) = \begin{cases} \left(a - \frac{1}{k}q_1 - \frac{1}{k}q_2 - c\right) q_i & \text{if } Q < ka \\ -cq_1 & \text{if } Q \geq ka \end{cases}$$

for $i = 1, 2$, where $q_1 \in S_1, q_2 \in S_2$ and $Q = q_1 + q_2$. The 2-player normal-form game $G := \{S_1, S_2, \pi_1, \pi_2\}$ is known as the **Cournot model of duopoly**.

It is clear that if $q_1 > ka - q_2$, then $\pi_1(q_1, q_2) = -cq_1 < 0$ so it cannot be an optimum. Therefore, we effectively only need to consider $q_1 \in [0, ka - q_2]$ and $q_2 \in [0, ka - q_1]$.

Proposition 1.1.17 ▸ Best Response in Cournot Model of Duopoly

Let $G := \{S_1, S_2, \pi_1, \pi_2\}$ be a Cournot model of duopoly, then for any $q_2 \in S_2$, the best response of the first firm is

$$q_1 := R_1(q_2) = \begin{cases} \frac{1}{2}(ka - q_2 - kc) & \textit{if } q_2 \leq k(a - c) \\ 0 & \textit{if } q_2 > k(a - c) \end{cases}.$$

Proof. Fix $q_2 \geq 0$ as the output of the second firm. Note that it suffices to consider the case where $\pi_1(q_1, q_2) = \left(a - \frac{1}{k}q_1 - \frac{1}{k}q_2 - c\right) q_1$. If $a - c - \frac{1}{k}q_2 < 0$, for any $q_1 > 0$, we have $\pi_1(q_1, q_2) < 0$ so $q_1 = 0$ is the best response. Otherwise, consider

$$\frac{\partial \pi_1}{\partial q_1} = -\frac{2}{k}q_1 + a - \frac{1}{k}q_2 - c = 0.$$

Solving the equation yields

$$q_1 = \frac{ka - q_2 - kc}{2}.$$

Therefore, the best response of the first firm is

$$R_1(q_2) = \begin{cases} \frac{1}{2}(ka - q_2 - kc) & \text{if } q_2 \leq k(a-c) \\ 0 & \text{if } q_2 > k(a-c) \end{cases}.$$

□

Remark. In a simplified case where $P(Q) = a - Q$ (i.e., $k = 1$), we have $a = q_0$ and so

$$R_1(q_2) = \begin{cases} \frac{1}{2}(a - q_2 - c) & \text{if } q_2 \leq a - c \\ 0 & \text{if } q_2 > a - c \end{cases}.$$

Using the above result, it is easy to compute the Nash equilibrium of the Cournot model by solving the system

$$\begin{cases} q_1^* & = \frac{1}{2}(ka - q_2^* - kc) \\ q_2^* & = \frac{1}{2}(ka - q_1^* - kc) \end{cases}.$$

Corollary 1.1.18 ▸ Nash Equilibrium of Cournot Model of Duopoly

In a Cournot model of duopoly, the Nash equilibrium is $q_1^* = q_2^* = \frac{ka-kc}{3}$.

Next, instead of **homogeneous product**, let us consider firms producing **differentiated products**. In this case, the Cournot model no longer makes sense because the sum of quantities demanded for differentiated products is meaningless.

Let c be the cost of production and a be the demand-side reservation price. We shall consider the model through the perspective of pricing. Suppose that the two firms set the prices for their products at p_1 and p_2 respectively. Take the demand for the first firm's product as an example. We may assume that the demand is linear in price and that $b > 0$ is a coefficient representing the substitutability of the first firm's product for the second firm's. Consider

$$Q(p_1) = q_0 - kp_1,$$

where the x-intercept is a, so we can compute

$$q_0 = ka.$$

Considering substitutability, we have

$$q_1(p_1, p_2) = ka - kp_1 + bp_2.$$

Definition 1.1.19 ▸ Bertrand Model of Duopoly

Let $S_1, S_2 := [0, \infty)$ be possible prices for two firms producing differentiated products. Let c and a be the cost of production and demand-side reservation price respectively. Assuming linear demand, then there exists positive constant k and b such that

$$\begin{aligned}\pi_i\left(p_1, p_2\right) &= q_i\left(p_1, p_2\right) p_i - c q_i\left(p_1, p_2\right) \\ &= \left(ka - kp_1 + bp_2\right)\left(p_i - c\right)\end{aligned}$$

for $i = 1, 2$, where $p_1 \in S_1, p_2 \in S_2$ are pricing strategies. The 2-player normal-form game $G := \{S_1, S_2, \pi_1, \pi_2\}$ is known as the **Bertrand model of duopoly**.

Via a similar process, we can compute the best responses and Nash equilibrium for the model.

Proposition 1.1.20 ▸ Nash Equilibrium in Bertrand Model of Duopoly

Let $G := \{S_1, S_2, \pi_1, \pi_2\}$ be a Bertrand model of duopoly, then for any $p_2 \in S_2$, the best response of the first firm is

$$R_1\left(p_2\right) = \frac{1}{2k}\left(ka + bp_2 + kc\right)$$

and the Nash equilibrium is

$$p_1^* = p_2^* = \frac{ka + kc}{2k - b}.$$

Remark. Note that this means that this model is sensible only if $b > 2k$.

We can use Bertrand model to analyse homogeneous product as well by setting $b = 0$. Assume that the consumers always buy from the firm which offers a lower price, then the demand for the first firm is

$$q_1\left(p_1, p_2\right) = \begin{cases} ka - kp_1 & \text{if } p_1 < p_2 \\ 0 & \text{if } p_1 > p_2 \\ \frac{ka - kp_1}{2} & \text{if } p_1 = p_2 \end{cases}.$$

The payoff can be computed as profit for the first firm by

$$\pi_1(p_1, p_2) = \begin{cases} (ka - kp_1)(p_1 - c) & \text{if } p_1 < p_2 \\ 0 & \text{if } p_1 > p_2 \\ \frac{1}{2}(ka - kp_1)(p_1 - c) & \text{if } p_1 = p_2 \end{cases}.$$

Proposition 1.1.21 ▸ Nash Equilibrium of Betrand Model for Homogeneous Product

The Nash equilibrium for the Bertrand model if both firms produce the same product is $p_1 = p_2 = c$ where c is the cost of production.

Proof. It suffices to consider the first firm. Notice that when $p_1 < c$ or $p_1 > a$, we have $\pi_1(p_1, p_2) < 0$, so it suffices to consider the strategy space as $[c, a]$, where we assume that $c \leq a$. We consider 3 cases. If $p_2 > \frac{a+c}{2}$, it is obvious that $p_1 = \frac{a+c}{2}$ is the best response. If $c < p_2 \leq \frac{a+c}{2}$, it is clear that

$$\sup \pi_1(p_1, p_2) = (ka - kp_2)(p_2 - c)$$

but there is no $p_1 \in [c, a]$ such that $\pi_1(p_1, p_2) = (ka - kp_2)(p_2 - c)$, and so there is no best response. If $p_2 = c$, then since $\pi_1(p_1, c) = 0$ for all $p_1 \in [c, a]$, we have $R_1(p_2) = [c, a]$. Observe that $R_1([c, a]) \cap R_2([c, a]) = \{(c, c)\}$, so the only Nash equilibrium is (c, c). $\square$

An important implication of this is that if consumers only buy from the cheaper supplier, then the firms have no choice but to produce at the cost without any profit.

Another classic claim in economics is that public goods tend to be under-supplied in a free market. This is summarised as the *Tragedy of the Commons*, which states the following problem:

> Suppose that there are n persons ($n \geq 2$) in a community. Let x_i be the amount of a public resource used by the i-th person and denote
>
> $$X := \sum_{i=1}^{n} x_i.$$
>
> For each unit usage of the resource, there is a fixed cost of c. The public resource has a total capacity of $X_{\max}$. When x units of the resources are used, each unit induces a utility of $v(x)$, where v is a strictly concave function for $x < X_{\max}$ and $v(x) = 0$ for $x \geq X_{\max}$. All the persons simultaneously choose how much of the public resource they wish to use.

If we model this as an n-player normal-form game, the strategy space for the i-th person will be $S_i := [0, X_{\max})$ and the payoff function is

$$u_i(x_i, \boldsymbol{x}_{-i}) := x_i v\left(\sum_{k=1}^{n} x_k\right) - c x_i.$$

Let $\boldsymbol{x}^* := (x_1^*, x_2^*, \cdots, x_n^*)$ be a Nash equilibrium, then clearly

$$\left.\frac{\partial u_i}{\partial x_i}\right|_{\boldsymbol{x}=\boldsymbol{x}^*} = v\left(\sum_{k=1}^{n} x_k^*\right) + x_i^* \left.\frac{\mathrm{d}v}{\mathrm{d}x_i}\right|_{x_i = x_i^*} - c = 0$$

for each $i = 1, 2, \cdots, n$. If we use

$$v(x) := \begin{cases} 0 & \text{if } x \geq X_{\max} \\ a - x^2 & \text{if } x < X_{\max} \end{cases}$$

for some $a > c$ as the utility function, we have

$$a - X^2 - 2x_i^* X - c = 0$$

for each $i = 1, 2, \cdots, n$. Intuitively, $x_1^* = x_2^* = \cdots = x_n^*$ because all payoff functions are the same, so

$$a - n^2 {x_i^*}^2 - 2n {x_i^*}^2 - c = 0$$

for each $i = 1, 2, \cdots, n$. Solving the equation yields $x_i^* = \sqrt{\frac{a-c}{n^2+2n}}$ for each $i = 1, 2, \cdots, n$ as the Nash equilibrium.

Next, define the *social welfare* as

$$w(x) = x v(x) - xc$$

where x is the total usage of the public resource. We have the following argument:

Proposition 1.1.22 ▸ The Tragedy of the Commons

Let $x^* < X_{\max}$ be the optimal usage for maximum social welfare and $(x_1^*, x_2^*, \cdots, x_n^*)$ be the Nash equilibrium, then

$$x^* < \sum_{i=1}^{n} x_i^*.$$

Proof. Denote $X^* := \sum_{i=1}^{n} x_i^* < X_{\max}$. Suppose on contrary that $x^* \geq X^*$, then we have

$$v(X^*) \geq v(x^*) \quad \text{and} \quad 0 > v'(X^*) \geq v'(x^*)$$

since v is strictly concave on $(-\infty, X_{\max})$. Note that $n \geq 2$, so $\frac{X^*}{n} < X^* \leq x^*$, which means that

$$\frac{X^*}{n} v'(X^*) \geq \frac{X^*}{n} v'(x^*) > x^* v'(x^*).$$

Therefore,

$$v(X^*) + \frac{X^*}{n} v'(X^*) > v(x^*) + x^* v'(x^*).$$

However, notice that since x^* achieves maximum social welfare, we have

$$w'(x^*) = v(x^*) + x^* v'(x^*) - c = 0.$$

For each $i = 1, 2, \cdots, n$, we also have $x_i^* = \frac{X^*}{n}$ and thus

$$v(X^*) + \frac{X^*}{n} v'(X^*) - c = 0.$$

Therefore, $v(X^*) + \frac{X^*}{n} v'(X^*) = v(x^*) + x^* v'(x^*)$, which is a contradiction. $\square$

The above argument precisely shows that at the Nash equilibrium, the usage of public resources always exceed what is socially optimal.

Game theory can also be used to examine bargaining. One such example is the negotiation for wages between firms and workers. Here, a mechanism called *final-offer arbitration* comes into play:

- The firm and the union simultaneously offer their desired wages, w_f and w_u, respectively.
- We may assume that $w_f < w_u$ due to the aims of the firm and the union.
- Suppose that an **arbitrator** has an ideal offer x and chooses the final offer by

  $$w = \begin{cases} w_f & \text{if } x < \frac{w_f + w_u}{2} \\ w_u & \text{if } x > \frac{w_f + w_u}{2} \end{cases}.$$

  In other words, the arbitrator chooses whichever is closer to his ideal expectation.
- Suppose that both the firm and the union **believe** that x is randomly distributed with probability density function f and cumulative distribution function F.

Denote x_m to be the median of x, i.e., $F(x_m) = \frac{1}{2}$. Notice that

$$P(w = w_f) = P\left(x < \frac{w_f + w_u}{2}\right) = F\left(\frac{w_f + w_u}{2}\right),$$
$$P(w = w_u) = 1 - F\left(\frac{w_f + w_u}{2}\right).$$

Let $\phi\left(w_f, w_u\right)$ be the expected wage settlement, then

$$\phi\left(w_f, w_u\right) = \mathbb{E}\left[w\right] = w_f F\left(\frac{w_f + w_u}{2}\right) + w_u\left(1 - F\left(\frac{w_f + w_u}{2}\right)\right).$$

Clearly, ϕ can be seen as a payoff for this two-player game.

Proposition 1.1.23 ▸ Nash Equilibrium of Final-Offer Arbitration

In a final-offer arbitration with ideal offer x distributed with probability density function f, cumulative distribution function F and median x_m, if w_f and w_u are the offers by the firm and the union respectively, then the Nash equilibrium is

$$w_f^* = x_m - \frac{1}{2f\left(x_m\right)} \quad \textit{and} \quad w_u^* = x_m + \frac{1}{2f\left(x_m\right)}.$$

Proof. Let $\phi\left(w_f, w_u\right)$ be the expected wage settlement, then $\left(w_f^*, w_u^*\right)$ are an Nash equilibrium if and only if

$$\phi\left(w_f^*, w_u^*\right) = \min_{0 \leq w_f < w_u^*} \phi\left(w_f, w_u^*\right) \quad \text{and} \quad \phi\left(w_f^*, w_u^*\right) = \max_{w_u > w_f^*} \phi\left(w_f^*, w_u\right).$$

Note that this implies that w_f^* is a local minimum of $w_f \mapsto \phi\left(w_f, w_u^*\right)$. Therefore, w_f^* must be a stationary point and

$$\left.\frac{\partial \phi}{\partial w_f}\right|_{w_f = w_f^*} = 0.$$

Note that

$$\begin{aligned} \frac{\partial \phi}{\partial w_f} &= F\left(\frac{w_f + w_u}{2}\right) + \frac{w_f}{2} F'\left(\frac{w_f + w_u}{2}\right) - \frac{w_u}{2} F'\left(\frac{w_f + w_u}{2}\right) \\ &= F\left(\frac{w_f + w_u}{2}\right) + \frac{w_f - w_u}{2} f\left(\frac{w_f + w_u}{2}\right). \end{aligned}$$

Therefore,

$$F\left(\frac{w_f^* + w_u^*}{2}\right) + \frac{w_f^* - w_u^*}{2} f\left(\frac{w_f^* + w_u^*}{2}\right) = 0.$$

Similarly, w_u^* is a local maximum of $w_u \mapsto \phi\left(w_f^*, w_u\right)$ and so

$$1 - F\left(\frac{w_f^* + w_u^*}{2}\right) + \frac{w_f^* - w_u^*}{2} f\left(\frac{w_f^* + w_u^*}{2}\right) = 0.$$

Therefore, we have

$$1 - F\left(\frac{w_f^* + w_u^*}{2}\right) = F\left(\frac{w_f^* + w_u^*}{2}\right)$$

and so $F\left(\frac{w_f^*+w_u^*}{2}\right) = \frac{1}{2}$. This means that $x_m = \frac{w_f^*+w_u^*}{2}$ and so

$$\frac{1}{2} + \frac{w_f^* - w_u^*}{2} f\left(x_m\right) = 0.$$

Therefore, $w_f^* - w_u^* = -\frac{1}{f(x_m)}$. Consider the system

$$\begin{cases} w_u^* - w_f^* &= \frac{1}{f(x_m)} \\ \frac{w_u^*+w_f^*}{2} &= x_m \end{cases}.$$

Solving the system yields

$$w_f^* = x_m - \frac{1}{2f\left(x_m\right)} \quad \text{and} \quad w_u^* = x_m + \frac{1}{2f\left(x_m\right)}.$$

□

1.2 Dynamic Games of Complete and Perfect Information

1.2.1 Backward Induction

In static games, the players move independently and the strategy of one player will not affect the context in which another player would make decisions. However, there are many games where this is not true. For example, in a turn-based game, the sequence in which the players move makes a significant impact on the strategy of each player. Furthermore, such games consist of multiple stages and so a player has to observe the past actions in order to predict the best decision. In this case, it makes more sense to talk about the "moves" of each player during his or her turn.

Definition 1.2.1 ▸ Action Space

In an n-player dynamic game, the **action space** of the i-th player is the set of all possible moves of the player during the turn, denoted by A_i.

Note that now, the extent to which a player can observe the past is important as this impacts how much information the player can acquire to make a decision in his or her own profit.

We will first discuss the first case where the information about the past moves are fully available.

Definition 1.2.2 ▸ Dynamic Games of Complete and Perfect Information

A **dynamic game** of **complete** and **perfect** information is a game where

- the players move in sequence,
- all previous moves are observed before the next move is taken, and
- payoffs are common knowledge to all players.

Such games can be presented with a *game tree* consisting of game states after each possible sequence of moves. A game state can be modelled by an ordered n-tuple $(x_{k1}, x_{k2}, \cdots, x_{kn})$ denoting the payoffs off all players after k moves. In this respect, the initial state can be represented by $\mathbf{0}$. Let $G := \{S_1, S_2, \cdots, S_n, u_1, u_2, \cdots, u_n\}$ be such a dynamic game.

Let $\mathbf{0}$ be the root of the game tree. For each payoff vector $\boldsymbol{u}_{h-1}$ in height $h - 1$, the h-th player takes a move in the *feasible set* A_h to modify the payoffs to a new payoff vector $\boldsymbol{u}_h$ and create an edge $s_h := \boldsymbol{u}_{h-1}$-$\boldsymbol{u}_h$. Consider the leaves of the tree formed this way at height n. Note that every strategy is uniquely represented as the $\mathbf{0}$-$\boldsymbol{u}$ path for some leaf $\boldsymbol{u}$, where the projection $\pi_i(\boldsymbol{u}) = u_i(s_1, s_2, \cdots, s_n)$ is the payoff corresponding to the strategy.

Definition 1.2.3 ▸ Game Tree

A **game tree** is a rooted tree representing a game. The vertices of a game tree are known as **decision nodes** and the edges represent the strategies of the players. Each leaf of the tree is known as a **terminal node** and is associated with a payoff tuple.

Contrary to perfect information, imperfect information occurs when some (not necessarily all) players of a game do not know the entire history of other players' moves during their turn.

The optimal strategy of a dynamic game of complete and perfect information can be obtained from a recursive algorithm. Suppose that player 1 chooses the move a_1. For the second player, he would then observe the previous move and choose

$$R_2(a_1) := \operatorname*{argmax}_{a_2 \in A_2} u_2(a_1, a_2)$$

as his *best response*. However, since player 1 knows about this fact, he would start the game by taking

$$a_1^* \in \operatorname*{argmax}_{a_1 \in A_1} u_1(a_1, R_2(a_1))$$

as his move. We say that $(a_1^*, R_2(a_1^*))$ is a *backward induction outcome* of the game. However, note that if $a_2^* = R_2(a_1^*)$, in general $u_1(a_1^*, a_2^*)$ might not be maximised. This difference from Nash equilibrium is exactly because of the fact that the players no longer take moves simultaneously, so the second player would gain additional information when making a decision.

Theorem 1.2.4 ▸ Backward Induction

Consider an n-player dynamic game with complete and perfect information. Define

$$b_j\left(b_1\right) := R_j\left(b_1, b_2, b_3, \cdots, b_{j-1}\right)$$

as the best response of the j-th player given the first action b_1 in the game, then the backward induction outcome is given by

$$a_1^* \in \operatorname*{argmax}_{a_1 \in A_1} u_1\left(a_1, b_2\left(a_1\right), b_3\left(a_1\right), \cdots, b_n\left(a_1\right)\right)$$

and $a_i^* := b_i\left(a_1^*\right)$ for $i = 1, 2, \cdots, n$.

In an alternative perspective, a static game can be viewed just as a dynamic game such that each player has completely no information about the other players' moves. Therefore, it is clear that we can find a generalised way to represent both types of games.

Definition 1.2.5 ▸ Extensive-Form Representation of Games

An **extensive-form representation** of a game specifies

1. the players in the game;
2. when each player takes a move;
3. each player's strategy space;
4. each player's knowledge of the game;
5. the payoffs for each possible combination of strategies.

The term "each player's knowledge of the game" can be a bit vague. To make it more precise, we represent a game with a game tree. Note that when a player takes the turn, he or she might be at one among many possible nodes in the game tree. If there is perfect information, the player would be able to trace the game's historical moves from the root and deduce exactly which node he or she is currently at. On the other hand, if the information is imperfect, the player will not be able to distinguish between all the possible nodes.

Definition 1.2.6 ▸ Information Set

An **information set** for a player is a set of decision nodes in a game such that

- all nodes in the set belong to the same player at the same turn, and
- the player has no information to deduce which node in the set they are currently

at.

Another way to see an information set is to view it as a set of decision nodes such that when the player is currently in the set, he or she does not know which decision node is the predecessor to his or her current node.

Remark. Intuitively, when the information is perfect, the player's information set is a singleton. Hence, a game has imperfect information if and only if there exists some non-singleton information set.

Every n-player extensive-form game can be represented as a game tree, such that the tree's decision nodes are partitioned into n subsets, each belonging to a distinct player. The nodes belonging to the same information set are joined with dotted lines by convention when drawing the tree.

In the case of a multi-stage game, each stage can be seen as a subgame, which can be formalised as follows:

Definition 1.2.7 ▸ Subgame

A **subgame** in an extensive-form game is a game represented by a subtree rooted at an internal decision node belonging to a singleton information set of the game tree such that for each decision node n in the subgame, all other decision nodes belonging to the information set of n are also in the subgame.

Notice that via extensive-form games, both static and dynamic games can be represented in the same way. Therefore, this implies that dynamic games also have Nash equilibria!

However, note that the "strategies" of a player in a dynamic game would depend on the other player's moves, so in this case, a strategy is then a **complete plan** consisting of all possible responses to previous moves in the game.

Definition 1.2.8 ▸ Strategy in Dynamic Games

A **strategy** in a dynamic game is a function that maps a set of past moves to a move of the current player.

Note that the payoff in a dynamic game is computed based on a **sequence of moves** instead of a set of strategies. Therefore, given a set of strategies of all players, we should first transform it into the corresponding sequence of moves.

Definition 1.2.9 ▸ Payoff in Dynamic Games

Let $\boldsymbol{s} := (s_1, s_2, \cdots, s_n)$ be a collection of strategies in an n-player dynamic game. Define $a_i : \prod_{j=1}^{n} S_j \to A_i$, where S_i is the strategy space and A_i is the set of all possible moves for the i-th player, such that

$$a_i(\boldsymbol{s}) = s_i(a_1(\boldsymbol{s}), a_2(\boldsymbol{s}), \cdots, a_{i-1}(\boldsymbol{s})).$$

The **payoff** for $\boldsymbol{s}$ is defined as

$$\widetilde{\boldsymbol{u}}(\boldsymbol{s}) := \boldsymbol{u}(a_1(\boldsymbol{s}), a_2(\boldsymbol{s}), \cdots, a_n(\boldsymbol{s})).$$

With the above definition, we can "collapse" a game tree for a dynamic game into a normal-form representation and find the Nash equilibria.

Intuitively, if all players choose their best moves sequentially after observing the previous moves, the end result should correspond to a Nash equilibrium.

Proposition 1.2.10 ▸ Backward Induction Outcome Induces a Nash Equilibrium

Let $(a_1^*, R_2(a_1^*))$ be a backward induction outcome of a 2-player dynamic game of complete and perfect information, then (a_1^*, R_2) is a Nash equilibrium.

Proof. Consider

$$\begin{aligned}
\widetilde{u}_1(a_1^*, R_2) &= u_1(a_1^*, R_2(a_1^*)) \\
&= \max_{a_1 \in A_1} u_1(a_1, R_2(a_1)) \\
&= \max_{a_1 \in A_1} \widetilde{u}_1(a_1, R_2).
\end{aligned}$$

Therefore, a_1^* is a best response to R_2. Similarly, for any $s_2 \in S_2$, since

$$\begin{aligned}
\widetilde{u}_2(a_1^*, R_2) &= u_2(a_1^*, R_2(a_1^*)) \\
&= \max_{a_2 \in A_2} u_2(a_1^*, a_2) \\
&\geq u_2(a_1^*, s_2(a_1^*)) \\
&= \widetilde{u}_2(a_1^*, s_2),
\end{aligned}$$

we know that R_2 is also a best response to a_1^*. Therefore, (a_1^*, R_2) is a Nash equilibrium. □

However, a Nash equilibrium might not be a backward induction outcome. For example, if (s_1^*, s_2^*) are a Nash equilibrium, we have

$$s_1^* = a_1^* = \operatorname*{argmax}_{a_1 \in A_1} \widetilde{u}_1(a_1, s_2^*) = \operatorname*{argmax}_{a_1 \in A_1} u_1(a_1, s_2^*(a_1)).$$

but this does not guarantee that $a_1^* = \operatorname{argmax}_{a_1 \in A_1} u_1(a_1, R_2(a_1))$ since $s_2^* \neq R_2$ in general.

Similarly, for a 2-stage dynamic game with imperfect information, the first-stage Nash equilibrium (a_1^*, a_2^*) and the second-stage Nash equilibrium $(a_3(a_1, a_2), a_4(a_1, a_2))$ given (a_1, a_2) induce a Nash equilibrium (a_1^*, a_2^*, a_3, a_4) in the normal-form representation.

However, in such cases, a Nash equilibrium might not lead to an optimal solution. To rule out these bad equilibria, we need a stronger definition.

Definition 1.2.11 ▸ Subgame Perfect Nash Equilibrium

A Nash equilibrium $(s_1^*, s_2^*, \cdots, s_n^*)$ is said to be **subgame perfect** if it constitutes a Nash equilibrium in every subgame.

Remark. It can be shown that if mixed strategies are allowed, every finite dynamic game of complete information contains at least one subgame perfect Nash equilibrium.

1.2.2 Economic Models

Using dynamic games, we are able to model economic activities just like static games.

Definition 1.2.12 ▸ Stackelberg Model of Duopoly

The **Stackelberg model of duopoly** studies a dominant firm 1 with a follower firm 2 whose production strategy spaces are $S_1, S_2 := [0, \infty)$ respectively. Assuming that the demand is linear in price the demand-side reservation price is a, then the profit of firm i is

$$\pi_i(q_1, q_2) = q_i(P(Q) - c)$$

where $Q = q_1 + q_2$ and

$$P(Q) = \begin{cases} a - \frac{1}{k}Q & \text{if } Q < a \\ 0 & \text{if } Q \geq a \end{cases}$$

for some positive constant k.

Note that

$$R_2(q_1) = \operatorname{argmax} \pi_2(q_1, q_2) = \begin{cases} \operatorname{argmax}\Big(q_2\left(a - \frac{1}{k}q_1 - \frac{1}{k}q_2 - c\right)\Big) & \text{if } q_1 + q_2 < a \\ \operatorname{argmax}(-cq_2) & \text{otherwise} \end{cases}.$$

Clearly, it suffices to consider the case where $q_1 + q_2 < a$. If $a - \frac{1}{k}q_1 - c < 0$, we have $q_2 = 0$ as the best response. Otherwise, we should find the maximum of the payoff via differentiation. This means that

$$R_2(q_1) = \begin{cases} \frac{1}{2}(ka - q_1 - kc) & \text{if } q_1 < k(a-c) \\ 0 & \text{otherwise} \end{cases}.$$

Consider

$$\pi_1(a_1, R_2(a_1)) = \begin{cases} q_1\left(a - \frac{1}{k}q_1 - \frac{ka - q_1 - kc}{2} - c\right) & \text{if } q_1 < k(a-c) \\ q_1\left(a - \frac{1}{k}q_1 - c\right) & \text{if } a - c \leq q_1 < a \\ -cq_1 & \text{if } q_1 \geq a \end{cases}.$$

Proposition 1.2.13 ▸ Best Response of the Dominant Firm in Stackelberg Model

In Stackelberg model of duopoly, the best response of the dominant firm is

$$q_1^* := \frac{ka - kc}{2}.$$

Proof. Note that when $q_1 \geq k(a-c)$, the profit of the dominant firm is non-positive and strictly decreasing in q_1, so it suffices to find

$$q_1^* := \operatorname*{argmax}_{a_1 < k(a-c)} \pi(a_1, R_2(a_1)).$$

Solving $\left.\frac{\mathrm{d}\pi_1}{\mathrm{d}q_1}\right|_{q_1 = q_1^*} = 0$ yields

$$q_1^* := \frac{ka - kc}{2}.$$

□

Using the above result, the follower firm's production quantity can be computed to be

$$q_2^* = R_2(q_1^*) = \frac{1}{2}(ka - q_1^* - kc) = \frac{ka - kc}{4}.$$

The price at the market is therefore

$$P\left(q_1^* + q_2^*\right) = P\left(\frac{3ka - 3kc}{4}\right) = a - \frac{3k\left(a-c\right)}{4k} = c + \frac{a-c}{4}.$$

Furthermore, the profits for the two firms are

$$\pi_1\left(q_1^*, q_2^*\right) = \frac{k^2\left(a-c\right)^2}{8} \quad \text{and} \quad \pi_2\left(q_1^*, q_2^*\right) = \frac{k^2\left(a-c\right)^2}{16}.$$

We can compare this model with Cournot model, with the latter seen as a game where the second firm makes production decisions independently without observing what the first firm decides.

	Cournot	Stackelberg
(q_1^*, q_2^*)	$\left(\frac{ka-kc}{3}, \frac{ka-kc}{3}\right)$	$\left(\frac{ka-kc}{2}, \frac{ka-kc}{4}\right)$
P	$c + \frac{a-c}{3}$	$c + \frac{a-c}{4}$
(π_1^*, π_2^*)	$\left(\frac{k^2(a-c)^2}{9}, \frac{k^2(a-c)^2}{9}\right)$	$\left(\frac{k^2(a-c)^2}{8}, \frac{k^2(a-c)^2}{16}\right)$
$\pi_1^* + \pi_2^*$	$\frac{2k^2(a-c)^2}{9}$	$\frac{3k^2(a-c)^2}{16}$

Surprisingly, knowing more information causes the second firm to obtain a lower profit in Stackelberg model as compared to Cournot model. This can be explained by the fact that since the first firm knows that the second firm would observe their decision first before making a move, this piece of information will be utilised by the first firm to make a better decision, causing the second firm to become worse off.

Dynamic games can be very useful in modelling bargaining. A simple model is as follows:

Definition 1.2.14 ▸ Three-Period Bargaining Game

The **three-period bargaining game** describes the following procedures:

1. In period t, player t proposes $s_t(t) \in [0,1]$ for himself and $s_{3-t}(t) := 1 - s_t(t)$ for the other player.
2. In the first 2 periods, the other player can either
   (a) accept the offer and end the bargaining, or
   (b) continue to the next period.
3. In period 3, player 1 gets $\bar{s}_1$ and player 2 gets $\bar{s}_2$ for some fixed exogenous amounts $\bar{s}_1, \bar{s}_2 \in [0,1]$ with $\bar{s}_1 + \bar{s}_2 \leq 1$.

The payoff to player i at period t is given by $u_i(t) = \delta^{t-1} s_i(t)$ for some $\delta \in (0,1)$.

Remark. Here, the constant δ represents the decay in satisfaction over time and is known as the *discount factor*.

Let us first consider the case where both players can predict correctly the amounts they will receive in the next period.

Proposition 1.2.15 ▸ Best Prudent Strategy in Three-Period Bargaining

Suppose that in a three-period bargaining game, the players will receive (u_1, u_2) for some $u_1, u_2 \geq 0$ with $u_1 + u_2 \leq 1$ in period $t+1$, then the best strategy for player t in period t is to offer $s_{3-t}(t) = \delta u_{3-t}$ and the best strategy for player $3-t$ is to accept the offer.

Proof. Observe that player $3-t$ in period t will accept the offer if and only if

$$u_{3-t}(t) = \delta^{t-1} s_{3-t}(t) \geq \delta^t u_{3-t}.$$

Clearly, player t will offer $s_{3-t}(t) \leq \delta u_{3-t}$. If $s_{3-t}(t) = \delta u_{3-t}$, then player $3-t$ will accept the offer and player t receives $1 - \delta u3 - t$. Otherwise, player $3-t$ will reject the offer and continue to the next period, where player t receives

$$u_t \leq 1 - u_{3-t} < 1 - \delta u_{3-t}.$$

Therefore, the best strategy of player t is to offer δu_{3-t} and the other player will accept the offer. $\square$

Using the above fact, we can deduce player 2's action during the second period since the final settlement is common knowledge. This allows us to perform backward induction.

Proposition 1.2.16 ▸ Backward Induction Outcom of Three-Period Bargaining

In a three-period bargaining game, player 1 will offer

$$\begin{cases} s_1(1) &= 1 - \delta(1 - \delta \bar{s}_1) \\ s_2(1) &= \delta(1 - \delta \bar{s}_1) \end{cases}$$

and player 2 will accept the offer.

Proof. Suppose that the bargaining proceeds to the second period, then by Proposition 1.2.15, player 2 will offer

$$\begin{cases} s_1(2) &= \delta \bar{s}_1 \\ s_2(2) &= 1 - \delta \bar{s}_1. \end{cases}$$

If player 1 offers $s_2(1) < \delta(1 - \delta\bar{s}_1)$, then

$$u_2(s_2(1)) = s_2(1) < \delta(1 - \delta\bar{s}_1) = u_2(s_2(2)),$$

so player 2 will not accept the offer and so the final payoff of player 1 will be

$$u_1(s_1(2)) = \delta^2\bar{s}_1 < \delta^2\bar{s}_1 + 1 - \delta = u_1(s_1(1)).$$

Therefore, player 1 will offer

$$\begin{cases} s_1(1) & = 1 - \delta(1 - \delta\bar{s}_1) \\ s_2(1) & = \delta(1 - \delta\bar{s}_1) \end{cases}$$

and player 2 will accept the offer. □

We can extend this model by allowing the bargaining steps to recur indefinitely.

Definition 1.2.17 ▸ Infinite-Horizon Bargaining Game

An **infinite-horizon bargaining game** describes the following procedures of a dynamic game for $k \in \mathbb{Z}^+$:

1. In period $2k - 1$, player 1 proposes $s_1(2k-1) \in [0,1]$ for himself and $s_2(2k-1) := 1 - s_1(2k-1)$ for the other player.
2. In period $2k$, player 2 proposes $s_2(2k) \in [0,1]$ for himself and $s_1(2k) := 1 - s_2(2k)$ for the other player.
3. In each period, the other player either accepts the offer or continues to the next period.
4. The game ends when one player accepts the offer.

The payoff to player i at period t is given by $u_i(t) = \delta^{t-1}s_i(t)$ for some $\delta \in (0,1)$.

The intuition here is that the game is "memoryless", i.e., the subgame starting at odd periods are essentially identical because the players make offers alternately and the payoff functions form a geometric progression.

Denote the entire game by G. Suppose that $(\bar{s}_1, \bar{s}_2)$ are the **final offers** the players can receive in the backward induction outcome. It is clear that $\bar{s}_1, \bar{s}_2 > 0$ and $\bar{s}_1 + \bar{s}_2 \leq 1$.

Suppose that we have reached period 3. Let G' be the infinite-horizon bargaining game starting at period 3 with $(\bar{s}_1, \bar{s}_2)$ as the final offer. Therefore, the game G is equivalent to a three-period bargaining game with exogenous settlement $(\bar{s}_1, \bar{s}_2)$. By Proposition 1.2.16, we know that

$$\begin{cases} s_1(1) &= 1 - \delta(1 - \delta\bar{s}_1) \\ s_2(1) &= \delta(1 - \delta\bar{s}_1) \end{cases}$$

will be the offer of player 1, which player 2 will accept. Therefore, we have

$$\begin{cases} \bar{s}_1 &= 1 - \delta(1 - \delta\bar{s}_1) \\ \bar{s}_2 &= \delta(1 - \delta\bar{s}_1) \end{cases}$$

Solving the system yields $\bar{s}_1 = \frac{1}{1+\delta}$ and $\bar{s}_2 = \frac{\delta}{1+\delta}$.

1.2.3 Multi-stage Games

The above bargaining game can be generalised to any dynamic game with a "periodic" behaviour. The decision nodes contained within the same period can be seen as its own "mini-game" which is known as a *stage*.

Definition 1.2.18 ▸ Multi-stage Game

A **multi-stage game** is a dynamic game which can be partitioned into static **stage games**.

In a simplified form, a 4-player game consists of two stages. In the first stage, players 1 and 2 take moves simultaneously. In the second stage, players 3 and 4 take moves simultaneously. In this case, we can view the second stage as a *static sub-game* given additional information from the first stage.

In this case, players 3 and 4 will condition their strategies on the strategies of players 1 and 2 in the first stage. In an ideal case, for each pair of strategies (a_1, a_2) played in the first stage, the second stage has an Nash equilibrium $(a_3(a_1, a_2), a_4(a_1, a_2))$.

Definition 1.2.19 ▸ Subgame Perfect Outcome

In a 4-player two-stage game of imperfect information, a **subgame perfect outcome** is an Nash equilibrium of the first stage (a_1^*, a_2^*) such that $(a_3^*(a_1^*, a_2^*), a_4^*(a_1^*, a_2^*))$ are an Nash equilibrium of the second stage.

Such a structure can be used to represent **tariffs**.

Definition 1.2.20 ▸ Tariff Game

A **tariff game** describes two countries 1 and 2 where firm i at country i produces h_i for domestic market and e_i for export. The governments simultaneously choose tariff rates t_1 and t_2, after which the firms simultaneously choose production strategies (h_1, e_1) and (h_2, e_2).

With cost of production at c and reservation price at a, the usual computation yields a subgame perfect Nash equilibrium to be

$$\begin{aligned} t_1^* = t_2^* &= \frac{a-c}{3}, \\ h_1^* = h_2^* &= \frac{4(a-c)}{9}, \\ e_1^* = e_2^* &= \frac{a-c}{9}. \end{aligned}$$

We introduce another economic model using a two-stage game.

Definition 1.2.21 ▸ Bank Runs

Bank Runs is a model where

- 2 investors each deposit x dollars at a bank, which the bank invests into a project;
- if the project is forced to stop, the investors can recover a dollars;
- if the project completes, the investors can earn $b > a$ dollars;
- the investors can withdraw their money either before the project completes (stage 1) or after (stage 2).

By observation, it is clear that at stage 2, the unique Nash equilibrium is for both investors to withdraw to earn $\frac{b}{2}$ each. This means that the payoff for both not withdrawing at stage 1 is $\left(\frac{b}{2}, \frac{b}{2}\right)$.

One may check that both withdrawing and both not withdrawing are the pure Nash equilibria of stage 1, so by Definition 1.2.11, both are the Nash equilibria of the game.

When the stages of a multi-stage game is the same across the entire game, we arrive at a special case.

Definition 1.2.22 ▸ Repeated Game

A **repeated game** is a multi-stage game where each stage is the same static game G.

For *finitely repeated games*, we have the following intuitive result:

Proposition 1.2.23 ▸ Nash Equilibrium of Finitely Repeated Games

If the stage game in a finitely repeated game has a unique Nash equilibrium, then it is the unique subgame perfect Nash equilibrium of the repeated game.

Infinitely repeated games are a generalisation of infinite-horizon bargaining. In an infinitely repeated game, the players discount future payoffs at a common discount factor $\delta \in (0,1)$. Given any sequence of strategies, we can compute the aggregate payoff of the strategies using the following definition:

Definition 1.2.24 ▸ Present Value

Let π_i be the payoff obtained at the i-th stage in an infinitely repeated game with discount factor $\delta \in (0,1)$, then the **present value** of the payoffs is defined as

$$\sum_{i=1}^{\infty} \delta^{i-1} \pi_i.$$

Suppose that the stage game of an infinitely repeated game is the Prisoner's Dilemma game with two actions R and L and the following payoff matrix:

$$\begin{array}{cc} & \begin{array}{cc} \mathrm{L} & \mathrm{R} \end{array} \\ \begin{array}{c} \mathrm{L} \\ \mathrm{R} \end{array} & \begin{bmatrix} (1,1) & (5,0) \\ (0,5) & (4,4) \end{bmatrix}. \end{array}$$

There are two typical strategies for the game:

1. **Non-cooperative Strategy** plays "L" in all stages and is denoted by NC.
2. **Trigger Strategy** plays "R" in the first stage and plays "L" if at least one player did L in the previous stage. This strategy is denoted by T.

Remark. The trigger strategy will R until the other player decides to L, where the action will switch to "L" forever. If both players adopt the trigger strategy, the game will continue by both co-operating forever.

Referring back to the analysis of Prisoners' Dilemma, we have the following result:

Proposition 1.2.25 ▸ Non-cooperative Strategy Is a Nash Equilibrium

(NC, NC) is always a Nash equilibrium.

Proof. Without loss of generality, assume that player 1 plays NC. If player 2 choose to R, his payoff decreases, and so player 2 will choose to L in every stage, and so (NC, NC) is a backward induction outcome and a Nash equilibrium by Proposition 1.2.10. □

Intuitively, whether the trigger strategy is worth playing depends on how much value it provides when switching to not co-operating.

Proposition 1.2.26 ▸ Condition for Trigger Strategy to Be a Nash Equilibrium

(T, T) is a Nash equilibrium if and only if $\delta \geq \frac{1}{4}$.

Proof. It suffices to show that $T \in R_2(T)$ if and only if $\delta \geq \frac{1}{4}$. Without loss of generality, suppose that player 1 adopts T. We consider 2 cases. If at least one player choose to L in the previous stage, then player 1 will choose to L forever starting in the current stage. Therefore, player 2 has to L forever as well.

If both player R in the previous stage or the current stage is the initial stage, player 1 will choose R. If player 2 chooses L, then player 1 will L from the next stage onwards. By the previous argument, player 2 will choose L from the next stage onwards as well. Therefore, player 2's payoff from this stage onwards is

$$5 + \sum_{i=1}^{\infty} \delta^i = 5 + \frac{\delta}{1-\delta}.$$

If player 2 chooses R, then the game will continue with (R, R) indefinitely and player 2's payoff is

$$4 \sum_{i=0}^{\infty} \delta^i = \frac{4}{1-\delta}.$$

Therefore, $T \in R_2(T)$ if and only if

$$\frac{4}{1-\delta} \geq 5 + \frac{\delta}{1-\delta},$$

i.e., $\delta \geq \frac{1}{4}$. □